

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION

WINTER SEMESTER, 2024-2025

DURATION: 3 HOURS

FULL MARKS: 150

CSE 4511: Computer Networks**Programmable calculators are not allowed. Do not write anything on the question paper.**

Answer all 6 (six) questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses. Appendix is present at the end of the question.

1. a) Consider a hypothetical IP addressing system where each IP address is 16 bits long, written as four 4-bit blocks separated by dots. For example:

1010.1100.0110.0001

3 + 3 +
3 + 4
(CO3)
(PO2)

CIDR notation is used. In each subnet, the first address is the network address and the last address is the broadcast address.

A university is allocated the address block:

1100.0000.0000.0000/5

The university must design an addressing scheme using Variable Length Subnet Masking (VLSM) under the following constraints: There are five departments with host requirements of 70, 50, 25, 20, and 10 hosts respectively. In addition, one subnet must be reserved for a Data Center that requires 60 hosts. All subnets must be allocated from the given address block without any overlap. The overall design objective is to minimize the total number of unused (wasted) addresses across all allocated subnets.

Answer the following, showing all intermediate steps and binary reasoning:

- i. Allocate the subnets using VLSM for each requirement. For each subnet, provide: Network address, Broadcast address, CIDR prefix
 - ii. Compute the total number of allocated addresses and the total number of unused addresses.
 - iii. A student proposes allocating subnets in ascending order of size. Explain why this approach may fail or lead to inefficient utilization, using this scenario as justification.
 - iv. Suggest a minimal change in allocation that would improve space utilization, and justify your answer.
- b) Explain the Internet approach to scalable routing with proper diagrams and reasoning.
- c) Assume a checksum of n bits is used for error detection in a packet-based communication system. You are given that the probability of an undetected error is:

4 + 3
(CO2)
(PO2)
10
(CO3)
(PO2)

$$P(\text{undetected error}) = \frac{1}{2^n}$$

Assume errors are random and independent across packets. A network link transmits 10^7 packets per day. Due to channel noise, 0.1% of the transmitted packets become corrupted. Now, if a 16-bit checksum is used, how many corrupted packets per day are expected to go undetected?

2. UDP and TCP are two of the most popular transport layer protocols we use in modern internet today. Even though UDP is unreliable, it has its own advantage of being faster than TCP.

a) Differentiate between message-oriented and byte-oriented protocol with an example.

msg oriented

byte oriented

5

(CO1)

(PO1)

b) Why UDP is considered faster than TCP? Give proper reasoning

5

(CO2)

(PO2)

c) Design a custom Application Layer Protocol for a chat application that uses UDP as its transport protocol. Provide a detailed specification of your packet structure and operational logic with justification.

15

(CO4)

(PO3)

3. a) Why do we use pseudoheader in TCP for checksum calculation? What are the contents of the pseudoheader?

3 + 2

(CO2)

(PO2)

b) TCP uses three-way handshake for connection establishment. Give a scenario where a two-way handshake (SYN and SYN+ACK) can be problematic.

5

(CO2)

(PO2)

c) In SCTP there is no half-closed association. Why?

5

(CO2)

(PO2)

d) State the objective of the send and receive windows in TCP, and then explain with diagrams how they fulfill the objective.

10

(CO2)

(PO2)

4. TCP is a reliable transport layer protocol. This means that an application program that delivers a stream of data to TCP relies on TCP to deliver the entire stream to the application program on the other end in order, without error, and without any part lost or duplicated.

a) How does TCP perform congestion control?

5

(CO2)

(PO2)

b) Consider a TCP connection between a sender and a receiver, where each segment size is 1000 bytes long, the initial sequence number of the sender is 1000, and the retransmission timeout (RTO) = 4 RTT. The sender transmits 8 segments back-to-back. Thus, the sequence numbers are:

3 × 5

(CO3)

(PO2)

1000, 2000, 3000, 4000, 5000, 6000, 7000, 8000

During the transmission, the segment with sequence number 3000 is lost, and the segment with sequence number 5000 is also lost. All other segments arrive at the receiver in the order in which they were transmitted. The retransmission timer is started for each segment at the time of its first transmission. ACK packets are assumed to never be lost. Any retransmitted segments arrive successfully and in the correct order. Answer the following questions:

i. What ACK number does the receiver send after receiving segment 1000?

- ii. At what point is fast retransmit triggered for segment 3000?
 - iii. After segment 3000 is successfully retransmitted and received, what ACK number does the receiver send?
 - iv. Explain whether segment 5000 will be retransmitted due to fast retransmit or timeout.
 - v. What is the final ACK number sent by the receiver after all data is successfully received?
- c) What are the differences between ACK and SACK?

5
(CO2)
(PO2)

5. (a) In SCTP, we use a four-way handshake for establishing a connection instead of a three-way handshake like TCP. What are the reasons behind this decision?

10
(CO2)
(PO2)

- (b) Why must the length field of a DATA chunk in SCTP be at least 17 bytes?

5
(CO2)
(PO2)

- c) A sender transmits DATA chunks with the following TSNs:

21, 22, 23, 24, 25, 26, 27, 28, 29, 30

2 +
3 + 5
(CO3)
(PO2)

The receiver responds with a SACK chunk containing:

Cumulative TSN = 23

Gap Ack Blocks = (26-28), (30-30)

Assume: No prior retransmissions have occurred. All chunks were sent exactly once. Standard SCTP SACK processing rules apply. Now, Answer the following questions:

- i. Which chunks are missing at the receiver?
- ii. Which chunks should be retransmitted immediately by the sender?
- iii. Explain why the SCTP retransmission queue has priority over new data chunks

5. You have developed a website at www.alliutians.com, a social media platform for iutians only. Now, to get monetary value from the website, you have introduced ads from AdX (www.adx.com). To give each user a unique and better browsing experience, you provide ads that align with their browsing behavior. For the simplicity of the question, suppose ads from AdX (www.adx.com) are used by many websites as well.

- a) With a diagram, explain how you can be successful in showing targeted ads. You can assume any random website the user visits is monetized by Adx (www.adx.com)

10
(CO4)
(PO3)

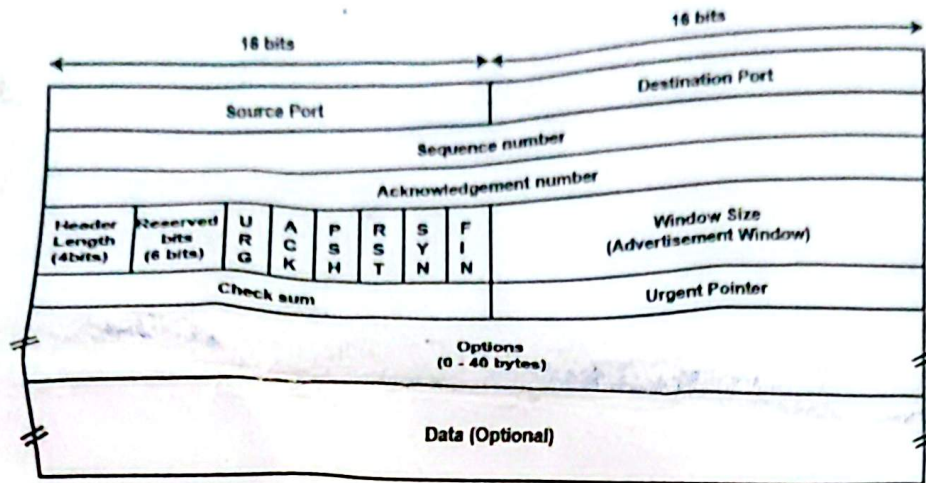
- b) Suppose your website consists of one base HTML file and 8 embedded objects (images/scripts/etc). Assume the round-trip time (RTT) between the client and the server is 100 ms. Ignore DNS lookup time and assume request/response transmission times are negligible compared to RTT. Compute the total time from the moment the client starts until all objects are downloaded in three scenarios. Show steps and numerical results (in milliseconds) for the following scenarios:

3 +
3 + 4
(CO3)
(PO2)

- 1. Non-persistent HTTP (no parallel connections)
- 2. Persistent HTTP with no pipelining (requests for embedded objects are made sequentially after the base HTML is received).
- 3. Persistent HTTP with pipelining (after receiving the base HTML, the client immediately sends requests for all 8 embedded objects back-to-back; server responds to these requests).

Appendix for CSE 4511

A Reference: Header & Packet Formats



TCP Header

Figure 1: TCP Header Format

Type=3	Chunk Flags	Chunk length
Cumulative TSN Ack		
Advertised Receiver window credit		
Number of GapACK Block		Number of Dup TSN
Gap ack block #1 start		Gap ack block #1 End
...		
Gap ack block #N start		Gap ack block #N End
Duplicate TSN 1		
...		
Duplicate TSN X		

Figure 2: SCTP SACK Chunk Format